

PRODUCT INFORMATION

Pressure Controller

High-responsiveness, high-accuracy
standard pressure generator



PC-4660 Pressure Controller

■ Features

- Control response: About 1.5 s (When the internal volume is 80mL) (20% of F.S.)
- Control accuracy: $\pm 0.05\%$ of F.S.
- Special control produces fewer overshoots.
- Also controls vacuum pressure and compound pressure.

■ Applications

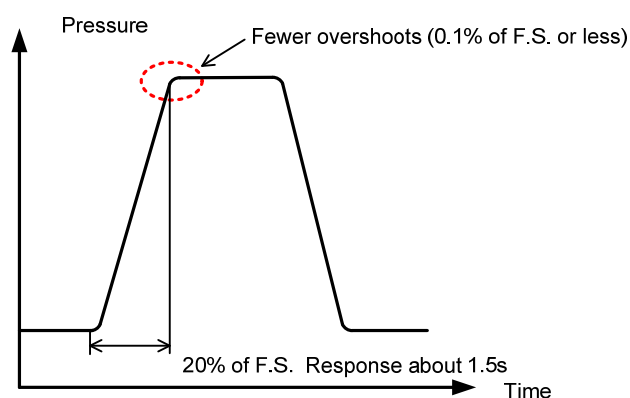
- Accuracy test on semiconductor pressure sensors
- Calibration of analog pressure gauges (such as bourdon tube gauges)
- Pressure switch operation test
- Vacuum valve characteristics test

The pressure controller is applicable only to sealed parts. Do not use it for leaking parts and/or parts that deform.

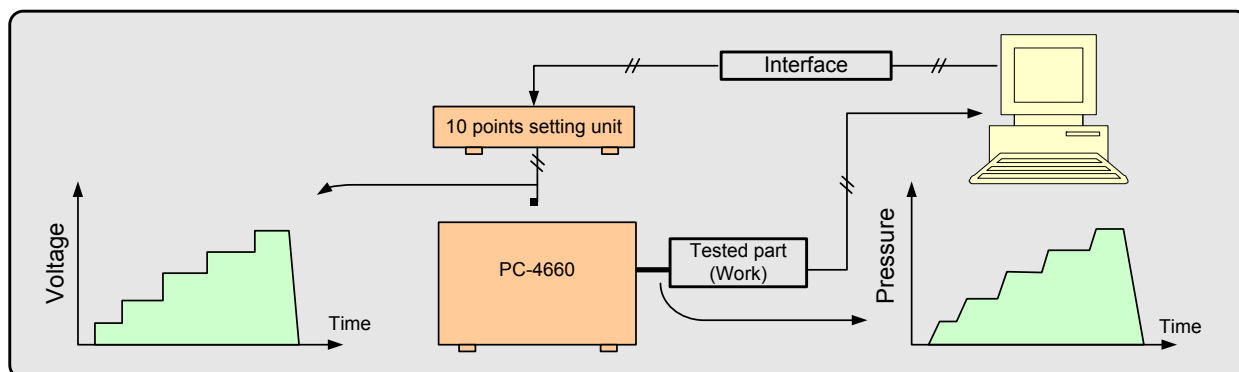
■ Overview

Special control valves enable more stable and faster pressure control. According to the internal settings or external input voltage, the air pressure is controlled highly accurately. At the same time, the value is digitally displayed.

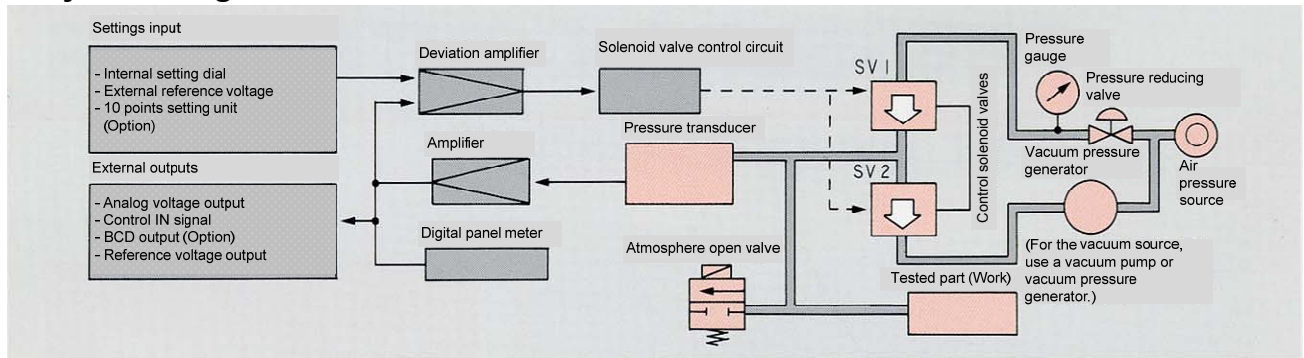
■ Pressure Control Graph



■ Pressure sensor accuracy test system



■ System Diagram

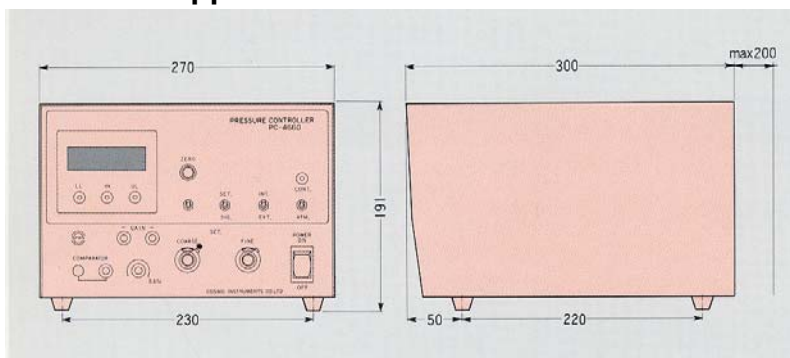


Pressure is controlled by SV1 (positive pressure control valve) and SV2 (vacuum pressure control valve). Switching of the valves is controlled at high-speed and accurately by the square-wave control current that is turned on and off at its natural frequency. Variation in the output pressure is always captured by the pressure sensor and fed back to the control circuit. When the tested part (Work) has a leak or deformation, the pressure cannot be controlled.

■ Specification

Range	0 to 1kPa, 0 to 2kPa 0 to 5kPa, 0 to 10kPa 0 to 20kPa, 0 to 50kPa, 0 to 100kPa, 0 to 200kPa, 0 to 500kPa, 0 to -100kPa
Parts to be tested (Work)	Sealed parts that have no leak, no deformation and no volume change. Larger internal volume takes longer control time.
Control accuracy	±0.05% of F.S.
Pressure sensor accuracy	±0.15% of F.S.
Pressure sensor temperature characteristics	Zero: ±0.03% /°C Span: ±0.03% /°C
Display	0 to ±19999 (4.5-digit display)
Port size	Rc1/8
Air pressure source	500kPa ≤ (When the pressure range is from 0 to 500kPa, the air pressure source should be at least 700kPa.) The pressure gauge, pressure reducing valve, and vacuum pressure generator are standard accessories of the controller.
Control response	When the pressure control range is 0 to 500 kPa Response: 20% of F.S., Approx.. 1.5 s (When the internal volume is 80 mL) Response: 90% of F.S., Approx.. 3 s (When the internal volume is 80 mL) When the vacuum control range is 0 to -70 kPa Response: 20% of F.S., Approx.. 1.5 s (When the internal volume is 80 mL) Response: 90% of F.S., Approx.. 3 s (When the internal volume is 80 mL) * For a vacuum greater than -70 kPa, longer response time than above is required. Contact Cosmo.
Power	AC100V±10%
Temp	5 to 40°C
Fluid	Air
Option	BCD output (open collector or TTL). Compound pressure, Auto-zero
Weight	Approx.. 12kg
Peripherals	10 points setting unit

■ External Appearance



■ I/O Signals

Pin name	Pin no.	Spec.
Analog output	18,19	0 to 5V DC
IN signal output	12,13	Output of contact point a
Reference voltage output	16,17	5.1V DC
External setting input	18,20	0 to 5V DC
Start / Stop signal input	14,15	When the signal is connected, control stops.

* Pins 16 and 18 are COM pins.

* The contents in this Product Information are as of October 2016. The information is subject to change without prior notice.

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